Problem 25.19

An n-year continuous annuity that pays at a rate of \$748 per year has a present value of \$10,000 when using an interest rate of 6%, compounded continuously. Determine n.

Solution

Since the interest rate is compounded continuously, the force of interest is

$$\delta = 0.06.$$

The present value of a continuous annuity payable at rate 748 per year for n years is

$$748\bar{a}_{\overline{n}|}.$$

We are given that this present value is 10,000, so

$$10000 = 748\bar{a}_{\overline{n}|}.$$

For a continuous annuity,

$$\bar{a}_{\overline{n}|} = \frac{1 - e^{-0.06n}}{0.06}.$$

Therefore,

$$\frac{10000}{748} = \frac{1 - e^{-0.06n}}{0.06}.$$

Multiplying by 0.06 gives

$$\frac{10000}{748}(0.06) = 1 - e^{-0.06n}.$$

Thus,

$$e^{-0.06n} = 1 - \frac{10000}{748}(0.06).$$

Taking natural logarithms,

$$-0.06n = \ln\left(1 - \frac{10000}{748}(0.06)\right).$$

Hence,

$$n = \frac{\ln\left(1 - \frac{10000}{748}(0.06)\right)}{-0.06}.$$

So

$$n = 27.0031784.$$

Therefore,

$$\boxed{n = 27.003}.$$